Recovering the unwritten recurrences of the table OEIS A236884

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Abstract

OEIS A236884 is a two-dimensional table $T(n, k)$ whose empirical recurrences are, for several of its lines, given only as an ORDER — "k=2: [order 31]" and the like — with the coefficients in a linked file rather than in the entry. Those conjectures can still be settled, and the recurrences recovered. A line of the table is a fixed-width or fixed-height array count, hence a walk count on finitely many vertices, hence satisfies some monic recurrence of order at most that number; Berlekamp–Massey on twice as many exact terms returns the MINIMAL one. Where its order equals the order the entry states — and where the same holds after the first few terms are dropped — any recurrence of that order the line satisfies has a characteristic polynomial that is a multiple of the minimal one and of equal degree, hence is that polynomial. This note settles column 2, column 3.

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1 The table and the unwritten conjectures

OEIS A236884 is "T(n,k) = Number of (n+1) X (k+1) 0..2 arrays with the maximum plus the upper median plus the lower median of every 2 X 2 subblock equal.". It is read by antidiagonals and begins

$$81, 211, 211, 537, 669, 537, 1523, 2111, \ \dots$$

The entry carries, among its empirical claims:

k=2: [order 31]
k=3: [order 63]

As of the "Last modified" line on the live entry (Jun 18 2022 18:00:45, revision 6) these are still recorded as empirical, and the recurrences themselves are not in the entry text.

2 Why a line of the table is a walk count

Fix the index that the line fixes. The entry defines $T(n, k)$ as a number of arrays of a shape determined by n and k, subject to a condition local to a bounded window of consecutive lines. Substituting the fixed index into the entry's own wording turns the definition into an ordinary one-parameter array count, and for such a count the admissible windows are the vertices of a finite digraph and an array is a walk. So the line is a walk count on S vertices, and by the Cayley–Hamilton theorem it satisfies a monic integer recurrence of order at most S.

3 Recovering the recurrences

Lemma 1. *A sequence known to satisfy some linear recurrence of order at most S is determined, with its minimal recurrence, by its first $2S$ terms; Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Column $k = 2$

Substituting the fixed index into the entry's wording gives an ordinary one-parameter array count, a walk on $S = 102$ vertices. Berlekamp–Massey on $2S = 204$ exact terms returns a minimal recurrence of order 31 — the order the entry states — with integer coefficients

$$(c_1, \ldots, c_{31}) = (14,\ -28,\ -459,\ 2258,\ 4418,\ -46337,\ 15855,\ 478245,\ -699780,\ -2810786,\ 6844631,\ 9089948,\ -$$

so that $a(m) = \sum_{i=1}^{31} c_i\, a(m-i)$ for every m. Removing the first $31 + 4$ terms and repeating gives order 31 again, which is what the identification below needs.

Column $k = 3$

Substituting the fixed index into the entry's wording gives an ordinary one-parameter array count, a walk on $S = 251$ vertices. Berlekamp–Massey on $2S = 502$ exact terms returns a minimal recurrence of order 63 — the order the entry states — with integer coefficients

$$(c_1, \ldots, c_{63}) = (16,\ 27,\ -1667,\ 4072,\ 76934,\ -341187,\ -2049421,\ 13078549,\ 33627314,\ -318556323,\ -306610$$

so that $a(m) = \sum_{i=1}^{63} c_i\, a(m-i)$ for every m. Removing the first $63 + 4$ terms and repeating gives order 63 again, which is what the identification below needs.

Theorem 2. *For each line above, the displayed recurrence holds for every index, and it is the recurrence the entry records.*

Proof. It holds by Lemma 1. For the identification, the entry asserts that the line satisfies SOME recurrence of the stated order, possibly only from some index on. Let q be its characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail. Then $q_{\min} \mid q$, and the second Berlekamp–Massey run — on the line with its first few terms removed — shows $\deg q_{\min}$ equals the stated order, which is $\deg q$. Both are monic, so $q = q_{\min}$. □

4 Verification

Three checks.

First, each line's model was compared against the entry's own published values for that line, taken from the antidiagonal DATA read in either orientation and from the "Table starts" block, and agrees at every available term. This is the only point at which reading the entry's English enters, and a misreading gives different counts.

Second, each recovered recurrence was evaluated directly on the model's values, with no Berlekamp–Massey involved, and holds at every index tested.

Third, each order was computed twice by independent routes — modulo a 61-bit prime, where every intermediate is a machine word, and again in exact rational arithmetic — and once more on the line with its first few terms dropped, which is what the identification requires. All agree.

References

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.